

Problem 3

Part (a)

First, we show that the probabilities of the two intervals are equal. By definition of the density, we have:

$$\begin{aligned}
 \mathbb{P}([x_0, \infty)) &= \int_{x_0}^{\infty} f(x) dx \\
 &= \int_0^{\infty} f(x_0 + u) du \quad (\text{Substitution: } u = x - x_0 \implies du = dx, x = u + x_0) \\
 &= \int_0^{\infty} f(x_0 - u) du \quad (\text{By symmetry}) \\
 &= \int_{x_0}^{-\infty} f(v)(-dv) \quad (\text{Substitution: } v = x_0 - u \implies dv = -du) \\
 &= \int_{-\infty}^{x_0} f(v) dv \\
 &= \mathbb{P}((-\infty, x_0])
 \end{aligned} \tag{1}$$

Second, we use the normalization property of probability densities (Lemma 1.5.3a). We know that the total probability over $\mathbb{R}$ must equal 1:

$$\begin{aligned}
 1 &= \int_{-\infty}^{\infty} f(x) dx \\
 &= \int_{-\infty}^{x_0} f(x) dx + \int_{x_0}^{\infty} f(x) dx \\
 &= \mathbb{P}((-\infty, x_0)) + \mathbb{P}([x_0, \infty)) \\
 &\stackrel{(\mathbb{P}(\{x_0\})=0)}{=} \mathbb{P}((-\infty, x_0]) + \mathbb{P}([x_0, \infty))
 \end{aligned}$$

Substituting (1) into this equation yields:

$$1 = 2\mathbb{P}([x_0, \infty)) \implies \mathbb{P}([x_0, \infty)) = \frac{1}{2} \tag{2}$$

Therefore, by (1) and (2), we have $\mathbb{P}([x_0, \infty)) = \mathbb{P}((-\infty, x_0]) = \frac{1}{2}$. □

Part (b)

For f to be a valid probability density function, it must satisfy three conditions: it must be Borel-measurable, normalized, and non-negative (Definition 1.5.12). We find the values of a and b to satisfy these properties:

1. **Borel-measurability:** The polynomial $x \mapsto ax^2 + b$ is continuous on $\mathbb{R}$ and therefore Borel-measurable. We can express $f(x) = (ax^2 + b)1_{(-1,1)}(x)$. Since the interval $(-1, 1)$ is open and thus a Borel set, the indicator function is measurable. Their product, $f(x)$, is therefore Borel-measurable.
2. **Normalization:** The total probability must equal 1. We evaluate the integral over the interval where $f(x)$ is non-zero (i.e., for $|x| < 1$):

$$\int_{\mathbb{R}} f(x) dx = \int_{-1}^1 (ax^2 + b) dx = \left[\frac{a}{3} x^3 + bx \right]_{-1}^1 = \left(\frac{a}{3} + b \right) - \left(-\frac{a}{3} - b \right) = \frac{2a}{3} + 2b$$

Setting this equal to 1 gives us the normalization condition:

$$\frac{2a}{3} + 2b = 1 \implies 2b = 1 - \frac{2a}{3} \implies b = \frac{1}{2} - \frac{a}{3}$$

Therefore $f(x)$ can be rewritten as:

$$f(x) = \begin{cases} ax^2 + \left(\frac{1}{2} - \frac{a}{3}\right), & \text{for } |x| < 1, \\ 0, & \text{otherwise.} \end{cases}$$

3. **Non-negativity:** We require $f(x) = ax^2 + b \geq 0$ for all $x \in (-1, 1)$ (as in the "otherwise" case, $f(x) = 0$, which is non-negative). In this interval, because $f(x)$ is a parabola, its absolute minimum occurs either at $x = 0$ (since $f'(x) = 2ax$ and the only critical point is $x = 0$) or at the boundaries ($x \rightarrow -1$ or $x \rightarrow 1$, which we denote $x \rightarrow \pm 1$ as the behaviour at both boundaries is identical due to the symmetry of the parabola). We must ensure $f(x) \geq 0$ at these points:

At $x = 0$:

$$f(0) \geq 0 \implies a(0)^2 + \left(\frac{1}{2} - \frac{a}{3}\right) \geq 0 \implies \frac{1}{2} - \frac{a}{3} \geq 0 \implies a \leq \frac{3}{2}$$

At the boundaries:

$$\lim_{x \rightarrow \pm 1} f(x) \geq 0 \implies a(\pm 1)^2 + \left(\frac{1}{2} - \frac{a}{3}\right) \geq 0 \implies \frac{2a}{3} \geq -\frac{1}{2} \implies a \geq -\frac{3}{4}$$

The function f is a valid density function if $a \in \left[-\frac{3}{4}, \frac{3}{2}\right]$ and $b = \frac{1}{2} - \frac{a}{3}$.

Part (c)

For f to be a valid probability density function, it must satisfy three conditions (Definition 1.5.12):

- **Borel-Measurability:** The function $(x, y) \mapsto x + y$ is continuous on $\mathbb{R}^2$ and therefore Borel-measurable. We can express $f(x, y) = (x + y)1_{(0,1)^2}(x, y)$. Since the set $(0, 1)^2$ is open (i.e., a Borel set), the indicator function is Borel measurable. Therefore $f(x, y)$ is the product of Borel-measurable functions and, thus, also Borel measurable.
- **Normalization:** The total probability must equal 1.

$$\begin{aligned} \int_{\mathbb{R}^2} f(x, y) dx dy &= \int_0^1 \int_0^1 (x + y) dx dy = \int_0^1 \left[\frac{x^2}{2} + yx \right]_{x=0}^{x=1} dy \\ &= \int_0^1 \left(\frac{1}{2} + y \right) dy = \left[\frac{y}{2} + \frac{y^2}{2} \right]_{y=0}^{y=1} = \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

- **Nonnegativity:** For $(x, y) \in (0, 1)^2$, we have $x > 0$ and $y > 0$, so $x + y > 0$. For all other (x, y) , $f(x, y) = 0$. Thus, $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$.

Therefore f is a valid density function of a probability measure $\mathbb{P}$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$.

Marginal densities:

Marginal density of X : For $x \notin (0, 1)$, $f_X(x) = \int_{\mathbb{R}} 0 \, dy = 0$. For $x \in (0, 1)$:

$$f_X(x) = \int_{\mathbb{R}} f(x, y) \, dy = \int_0^1 (x + y) \, dy = \left[xy + \frac{y^2}{2} \right]_{y=0}^{y=1} = x + \frac{1}{2}$$

Written compactly with an indicator function:

$$f_X(x) = \left(x + \frac{1}{2} \right) 1_{(0,1)}(x)$$

Marginal density of Y : The procedure is analogous to the marginal density of X :

$$f_Y(y) = \left(y + \frac{1}{2} \right) 1_{(0,1)}(y)$$